

Neural responses to binocular correlated and anticorrelated noise stimuli

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Abstract

The visual system encodes binocular signals via summing and differencing channels, yet the relative weighting of these pathways remains unclear. Here, we use Steady-State Visually Evoked Potentials (SSVEPs) to investigate the neural responses associated with summing and differencing mechanisms. In Experiment 1, we measured responses to binocular noise stimuli in 15 observers. Across the eyes, stimuli could oscillate (at 3Hz) from no correlation to perfect interocular correlation, from no correlation to perfect interocular anticorrelation (i.e., opposite interocular contrast), or no interocular correlation (i.e., the control condition). Responses to correlated stimuli showed a peak at the fundamental frequency (3Hz); however, we found no evidence of measurable responses to anticorrelation. In Experiment 2, we increased the sample size to 48 observers and measured SSVEPs in response to interocular correlation and anticorrelation while manipulating the degree of interocular correlation. The correlation between the left and right eye images could be ± 0 , 0.12, 0.24, 0.48, or 0.96. We found SSVEP responses at the fundamental frequency in both interocular correlation and anticorrelation conditions. Interocular correlation generated SSVEPs with a correlation of 0.48 or 0.96, while anticorrelation SSVEPs were only measurable at a correlation of -0.96. Signal-to-noise ratios were 1.15 times larger for the correlation condition than the anticorrelation condition. To determine the influence of individual differences on SSVEPs to interocular correlation, we collected psychophysical measures of binocular function and correlated them with neural responses. Two measures of binocularity were used: stereoacuity and binocular balance, which refers to the ratio of dominance duration in each eye during rivalry. We found only small associations between SSVEP magnitudes and psychophysical measures of binocularity (stereoacuity and binocular balance). We conclude that while neural responses to both correlation and anticorrelation can be recorded via SSVEPs, the latter requires high signal coherence and a large sample size.

Introduction

The visual system encodes binocular signals via summing and differencing channels, yet the relative weighting of these pathways remains unclear.

Here, we use Steady-State Visually Evoked Potentials (SSVEPs) to investigate the neural responses associated with summing and differencing mechanisms.

The studies presented in this manuscript are therefore twofold. First, we present findings from a pilot study where the data was collected as part of a larger experiment (Richard and Baker, 2025). In this study, observer SSVEPs were recorded to noise stimuli that oscillated in the interocular correlation

Experiment 1

Methods

Participants

Fifteen observers ($N_{male} = 4$; age range [19 34]), including authors BR and DHB, participated in this study. All observers had normal or corrected-to-normal visual acuity, as verified by a Snellen chart, and normal binocular vision, as verified by a Titmus test. Written informed consent was obtained from all participants, and experimental procedures were approved by the ethics committee of the Department of Psychology at the University of York.

Apparatus

All stimuli were presented using a gamma-corrected ViewPixx 3D display (VPixx Technologies, Canada) driven by a Mac Pro. Binocular separation with minimal crosstalk was achieved by synchronizing the display’s refresh rate with the toggling of a pair of Nvidia stereo shutter goggles using an infrared signal. The monitor refresh rate was set to 120 Hz; each eye updated at 60 Hz (every 16.67 msec). The display resolution was set to 1920×1080 pixels. A single pixel subtended 0.027° of visual angle (1.63 arc min) when viewed from 57 cm. The mean luminance of the display viewed through the shutter goggles was 26 cd/m^2 . EEG signals were recorded from 64 electrodes distributed across the scalp according to the 10/20 EEG system (Chatrian et al., 1985) in a WaveGuard cap (ANT Neuro, Netherlands). We monitored eye blinks with an electrooculogram consisting of bipolar electrodes placed above the eyebrow and the cheek on the left side of the participant’s face. Stimulus-contingent triggers were sent from the ViewPixx display to the amplifier using a parallel cable. Signals were amplified and digitized using a PC with the ASAlab software (ANT Neuro, Netherlands). All EEG data were imported into MATLAB (Mathworks, MA, USA) using components of the *EEGlab* toolbox (Delorme and Makeig, 2004) and then exported for subsequent offline analysis using R.

Stimulus Creation

Observer SSVEPs to interocular correlation and anticorrelation were measured with noise stimuli subtending 15° of visual angle on the retina (Figure 1). The visual noise stimuli were generated from the Fourier transform of a random matrix and assigning the DC a value of 0. The result is a flat ($\alpha = 0$) amplitude spectrum which was then low pass filtered with a rectangular function at 3 cycles/ $^\circ$. Interocular correlation and anticorrelation was generated by manipulating the phase spectrum of the noise image along a sinusoidal profile. Two phase matrices were constructed by sampling, from a uniform distribution, values between $-\pi$ and π while maintaining an odd-symmetric phase relationship to ensure conjugate symmetry. Interocular correlation was controlled by multiplying the two phase matrices by a sinusoidal function rectified between 0 and 1 and with a temporal frequency of 3Hz prior to adding to the original phase spectrum of the stimulus. In this way, we created two images with varying degrees of interocular correlation; the left and right eye images were perfectly correlated ($r = 1$) when the sinusoidal function was 0 and were uncorrelated ($r = 0$) when the sinusoidal function had a value of 1. The sign of pixels was reversed to create the anticorrelation images. All stimuli were generated using *MATLAB* (MathWorks, Natick, MA) and presented to observers using *Psychtoolbox* (Brainard, 1997; Kleiner et al., 2007; Pelli, 1997).

To aid with binocular fusion, stimuli were surrounded by a static binocular texture presented beyond the central 19° stimulus aperture. These textures were constructed by first low-pass filtering a white (amplitude $\propto 1/f^0$) noise pattern, dichotomizing its output into a binary image, and taking its phase spectrum. A second flat (amplitude $\propto 1/f^0$) was adjusted by multiplying each spatial frequency’s amplitude coefficient by f^{-1} to generate a pink amplitude spectrum (Hansen and Hess, 2006; Tadmor and Tolhurst, 1994). The pink amplitude spectrum and the phase spectrum of the binary image were rendered in the spatial domain by taking the inverse Fourier transform. The surrounding texture was fixed during a trial, but its phase structure was changed across trials.

Observer stereoacuity was measured by adding a depth cue to the noise stimuli used in the SSVEP experiment. The disparity signal was a horizontal sinusoidal wave within the noise stimulus that was added by filtering the noise image by a 2D Gaussian filter (Equation 1) in the Fourier domain, in a similar manner as Ding and Levi (2011).

$$G_L() = \exp - \left[\frac{(X - X_{D_i}^2)}{2\sigma^2} + \frac{(Y - Y_D)^2}{2\sigma^2} \right] \quad (1)$$

The horizontal disparity was determined by the value (X_{D_i}) for a given row (i) of the noise image.

Procedures

SSVEP Procedures

Steady-State Visually Evoked Potentials (SSVEPs) were recorded to stimuli that oscillated,

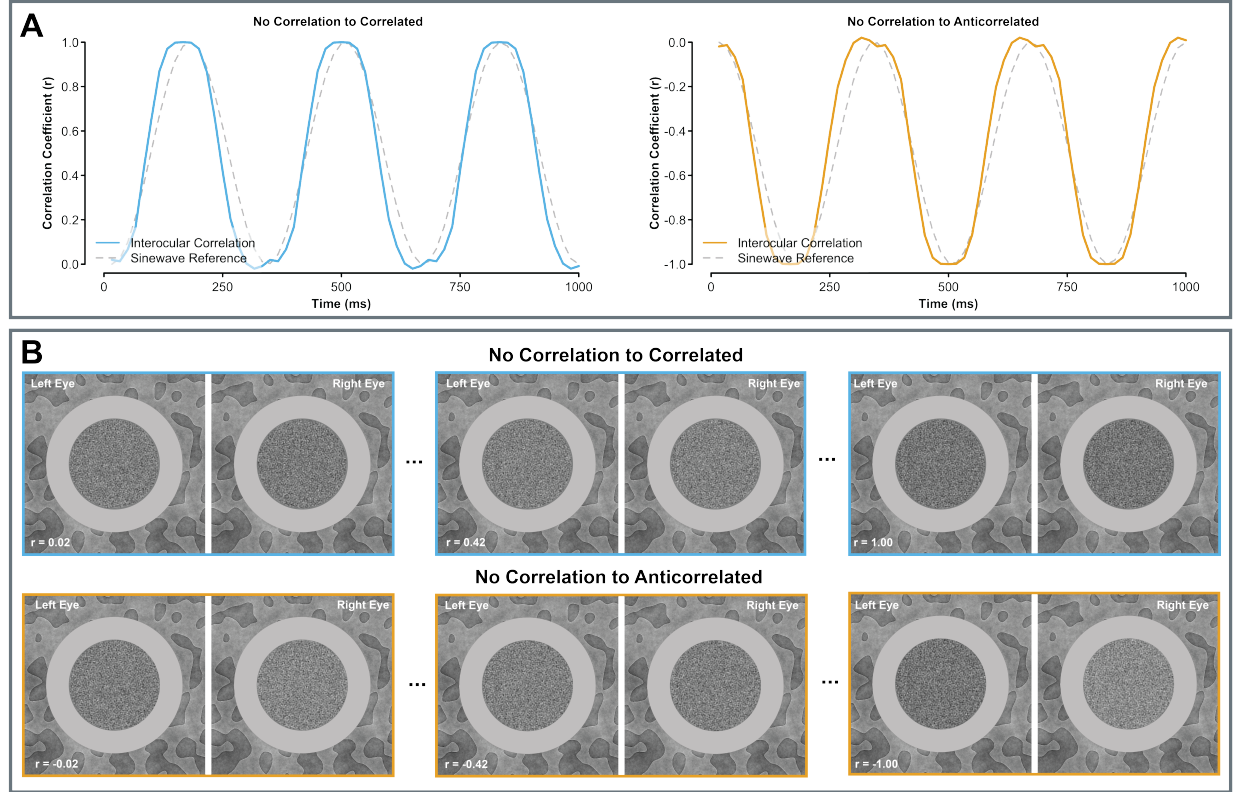


Figure 1: **A**. The sinusoidal oscillation of correlation strength between the left and right eye images. The blue line represents the oscillation for correlated images, while the mustard line that of anticorrelated stimuli. The dashed line is a sine (blue) or cosine (mustard) wave at 3Hz to serve as a reference. There are slight deviations between the correlation coefficient modulation and the reference sinewave. **B**. Example stimuli at three different correlation coefficients (± 0.00 , 0.42 , and 1.00) across the left and right eye. The top row shows stimuli that oscillate from uncorrelated to perfectly correlated while the bottom row shows stimuli that oscillate from uncorrelated to anticorrelated.

at a rate of 3Hz, from a correlation coefficient of 0 to 1 (correlated) and from a correlation coefficient of 0 to -1 (anticorrelated). A third, control, condition presented observers with the same phase manipulation but without any interocular correlation. These data were part of a larger experiment (Richard and Baker, 2025). The procedures reported here are those of the entire experiment. Observers completed 12 experimental conditions for a total of 144 trials (~40 minutes of EEG recording per participant) split into four experimental blocks. A trial lasted 15 seconds; the stimuli were presented on screen for 11 seconds followed by a screen with the central 19° set to mean luminance for 4 seconds. Participants completed all trials of an experimental block in a single sequence (9 minutes) and were given breaks between experimental blocks. was pseudo-randomized on each block. Participants did not receive explicit task instructions other than to fixate the marker in the center of the display and blink only during the blank period between stimulus presentations.

Psychophysical Procedures

Observer stereoacuity was measured using a 1-up, 3-down two-interval forced choice procedure (2-IFC) using the *PAL_AMUD_setupUD* and *PAL_AMUD_updateUD* functions from the Palamedes toolbox for MATLAB (Prins and Kingdom, 2018).

SSVEP Analysis

We used whole-head average referencing to normalize each electrode to the mean signal of all 64 electrodes (for each sample point). The EEG waveforms were Fourier transformed at each electrode for a 10-second window, beginning one second after the stimulus onset to avoid onset transients. The Fourier spectra were coherently averaged (i.e., retaining the phase information) across four occipital electrodes (*Oz*, *POz*, *O1*, and *O2*) and trial repetitions (see Figure 2). We then calculated signal-to-noise ratios (SNRs) by dividing the absolute amplitude in the signal bin (e.g., 3 Hz) by the mean of the absolute value of the ten adjacent bins (± 0.5 Hz in steps of 0.1 Hz). There were no differences in the amplitude of the adjacent, noise, bins across experimental conditions. Given that distributions of ratios (including SNRs) are inherently skewed, the median SNR was taken across all participants; the median is a more robust descriptor of central tendency in skewed distributions.

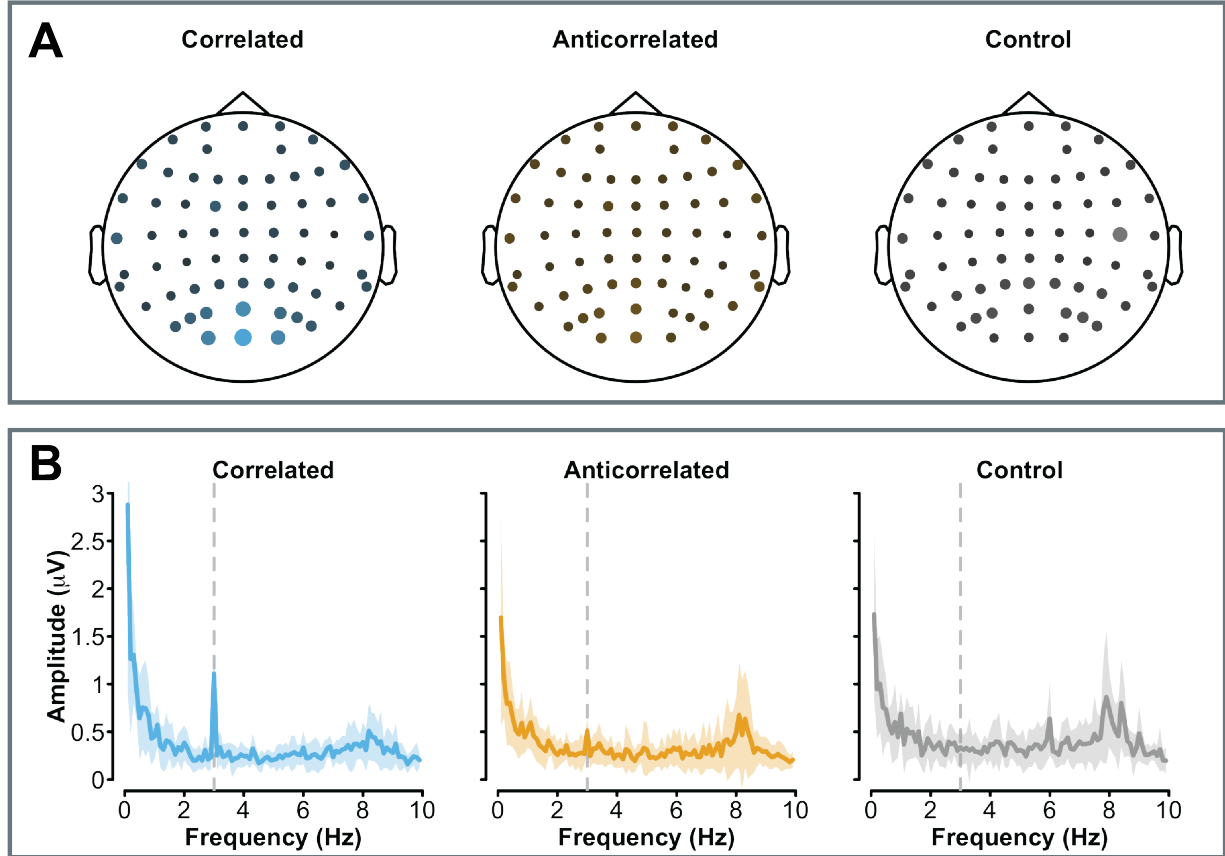
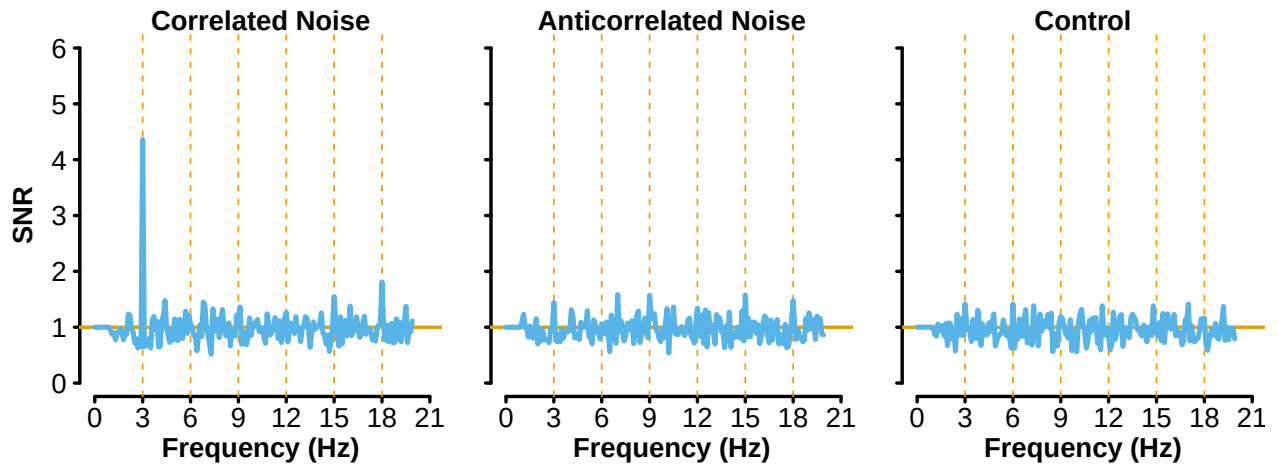
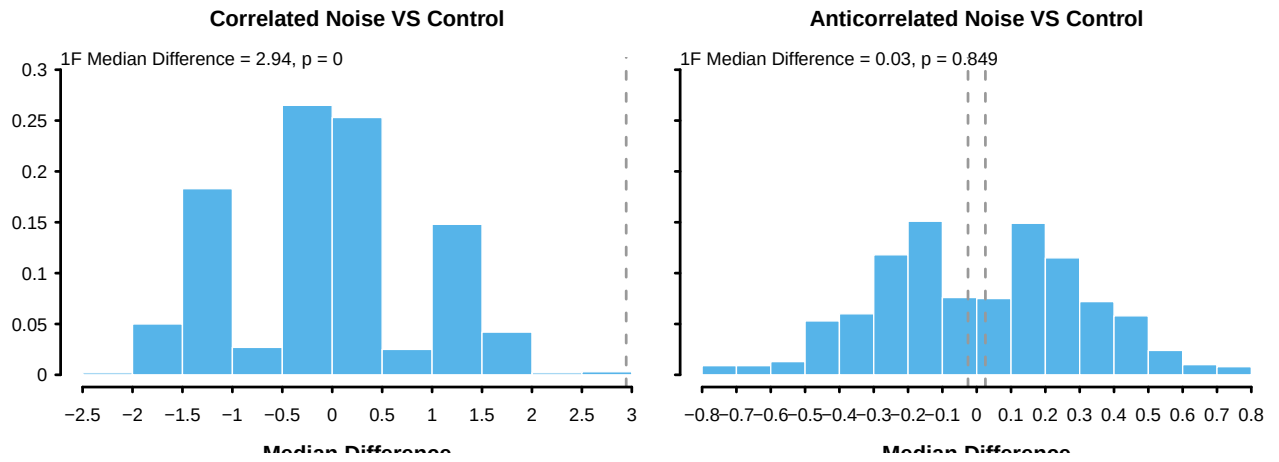


Figure 2: **A.** Head plots of the cross-participant average SSVEPs for the three experimental conditions reported in this experiment (correlated, anticorrelated, control). The size of marker represents the recorded amplitude at that electrode. Neural responses were largest at the occipital electrodes at the back of the head. This is most visible in the correlated condition, while SSVEPs were smaller in the anticorrelated condition. **B** The cross-participant average SSVEPs aggregated over the four occipital electrodes. As is shown in the headplots, SSVEPs at 3Hz were largest for the correlated condition. A small peak in amplitude can be seen at 3Hz in the anticorrelated condition. There was not SSVEP response at the occipital electrodes for the control condition.

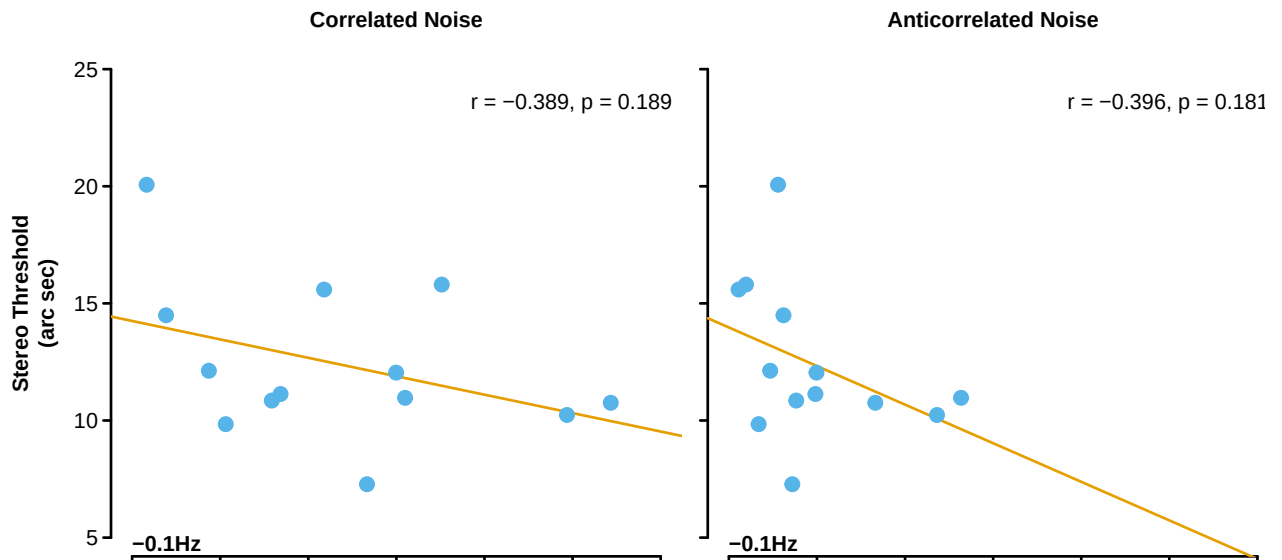
Results



Condition Comparisons

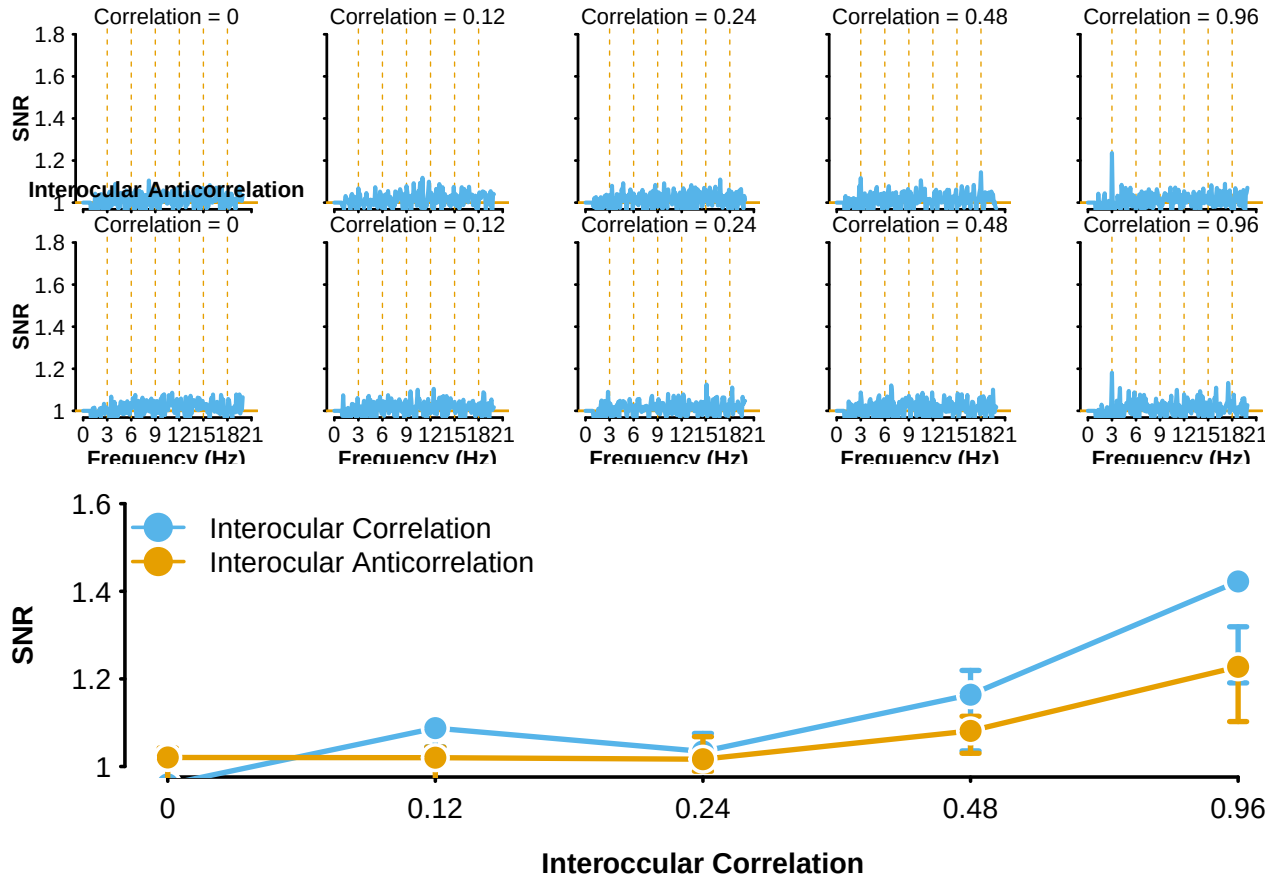


Correlation between SNRs and Stereothresholds

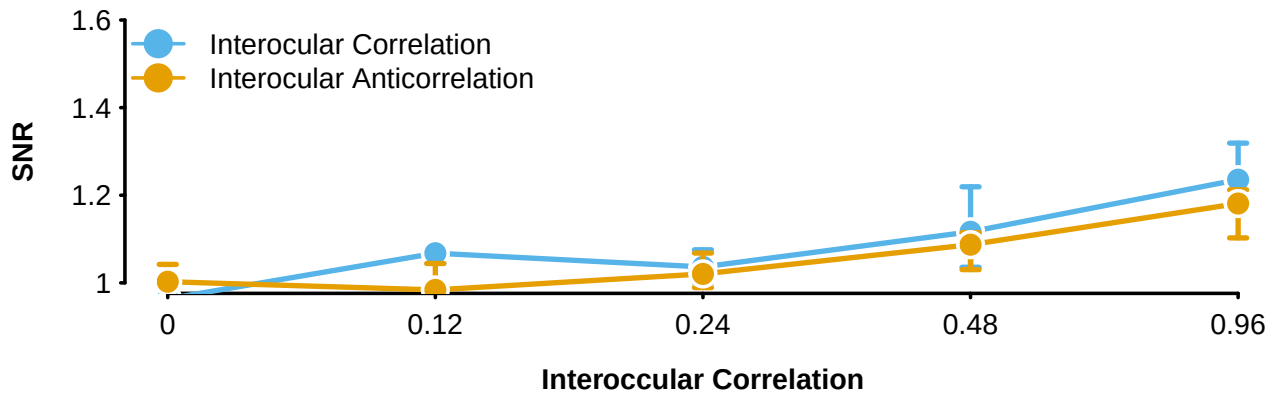


Experiment 2

Median SNRs

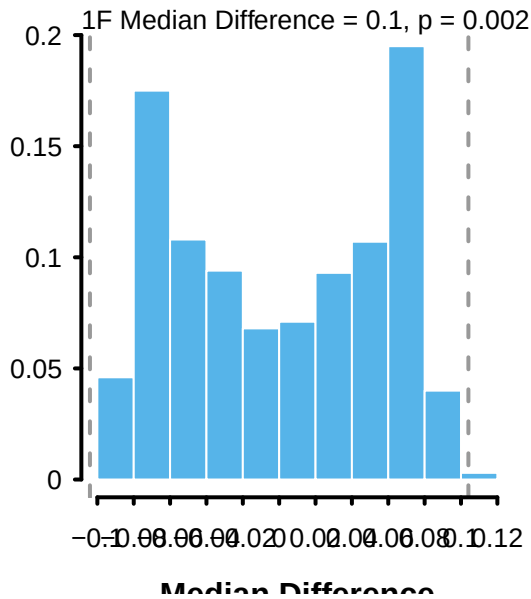


[1] 0.9398489 1.0661557 1.0174877 1.0764051 1.1587519



Compare the conditions to control

0.12 VS Control



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